

SF1811 Optimization

Lecture 3: The simplex method

Chapter 5

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The simplex method

Main question:

How does the simplex method move from one basic feasible solution to a better one?

Today:

- Improvement of the objective value.
- A feasible direction.
- The maximum feasible step.
- The basis change.
- The Tables and chairs problem.

$$Ax = b$$

$$\sum_{j=1}^{n} a_j x_j = b$$

a_j Column j of A

B, N column indices

N has dimension $n \times m$

B has dimension m

$$A = (A_B \; A_N) \quad x = \begin{pmatrix} x_B \\ x_N \end{pmatrix}$$

$$A_B x_B + A_N x_N = b \qquad x_N = 0 \qquad A_B x_B = b$$

A_B nonsingular $\Rightarrow$ $x_B = A_B^{-1} b$ Solve with A_B

Current basic feasible solution

Consider a linear program in standard form:

$$\begin{array}{ll} \text{minimize} & c^{\mathsf{T}} x \\ \text{subject to} & Ax = b, \\ & x \geq 0. \end{array}$$

$A \in \mathbb{R}^{m \times n}$

$\operatorname{rank}(A) = m$

For the current basis B,

$$x_N = 0, \qquad A_B x_B = b, \qquad x_B \geq 0.$$

Solve

Assumption

The current objective value is

$$z = c_B^{\mathsf{T}} x_B.$$

$z = (c_B^T \; c_N^T) \begin{pmatrix} x_B \\ x_N \end{pmatrix}$, $x_N = 0$

Can increasing a nonbasic variable improve the objective value?

Expressing the objective value

For any feasible solution,

$$A_B x_B + A_N x_N = b.$$

Let y satisfy

$$A_B^T y = c_B.$$

Then

$$\begin{aligned} c^T x &= c_B^T x_B + c_N^T x_N \\ &= y^T A_B x_B + c_N^T x_N \\ &= y^T (b - A_N x_N) + c_N^T x_N. \\ &= y^T b - y^T A_N x_N + c_N^T x_N \\ &= b^T y + (c_N^T - y^T A_N) x_N \end{aligned}$$

r_N^T

$x_N \geq 0$ allowed

$A_B x_B = b - A_N x_N$

A_B invertible

$x_B = A_B^{-1}(b - A_N x_N)$

$z = c_B^T x_B + c_N^T x_N$

$= c_B^T A_B^{-1}(b - A_N x_N) + c_N^T x_N$

$y^T = c_B^T A_B^{-1}$

$y = A_B^{-T} c_B$

$x_N = 0 \Rightarrow c^T x = b^T y$

$r_j = c_j - a_j^T y, \; j \in N$

jth column of A

Reduced costs

Define the vector of reduced costs by

$$r_N = c_N - A_N^T y, \qquad A_B^T y = c_B.$$

The objective value can then be written as

$$c^T x = b^T y + r_N^T x_N.$$

At the current basic feasible solution, $x_N = 0$.

- If $r_N \geq 0$, the current basic feasible solution is optimal.
- If $r_k < 0$ for some $k \in N$, increasing x_k may decrease the objective value.

The reduced cost r_k measures the change in objective value per unit increase of x_k.

Choosing an entering variable

Suppose that

$$r_k < 0 \qquad \text{for some } k \in N.$$

Choose x_k as the entering variable and set

$$x_k = t, \qquad t \geq 0,$$

while the other nonbasic variables remain zero.

The objective value becomes

$$z(t) = c_B^{\mathsf{T}} x_B + r_k t.$$

Since $r_k < 0$, the objective value decreases as t increases.

How must the basic variables change in order to preserve feasibility?

Maintaining the equations

Write the values of the basic variables as

$$x_B(t) = x_B + t\Delta x_B.$$

$x_B(t) = A_B^{-1}b - A_B^{-1}a_k t$

$x_B \quad \Delta x_B$

The equality constraints require

$$A_B x_B(t) + a_k t = b.$$

$\Delta x_B = A_B^{-1} a_k$

Solve $A_B \Delta x_B = a_k$

Since $A_B x_B = b$, it follows that

$$\boxed{A_B \Delta x_B = -a_k.}$$

x_B gives the current values of the basic variables.
Δx_B gives their change per unit increase of x_k.

The maximum feasible step

We want to increase t while maintaining

$$x_B(t) = x_B + t\Delta x_B \geq 0.$$

Only components for which

$$(\Delta x_B)_i < 0$$

restrict the step length.

Therefore,

$$t_{\max} = \min_{i:(\Delta x_B)_i<0} \frac{(x_B)_i}{-(\Delta x_B)_i}.$$

If the minimum is attained for $i = p$, the basic variable with index B_p reaches zero.

The entering index k replaces the leaving index B_p in the basis.

One simplex iteration

We can think of y and r as certificates of optimality.

Starting from a basic feasible solution:

1. Solve
$$A_B^T y = c_B$$
and compute $r_N = c_N - A_N^T y$.
2. If $r_N \geq 0$, stop: the current solution is optimal.
3. Otherwise choose $k \in N$ with $r_k < 0$.

Dantzig $\Rightarrow$ $\min_{j \in N} \{r_j\}$

4. Solve
$$A_B \Delta x_B = -a_k.$$
5. Determine $t_{\max}$ and the leaving index B_p.
6. Let k enter the basis and B_p leave the basis.

Tables and chairs: notation for the simplex method

For the simplex calculations, write

$$x_1 = T, \qquad x_2 = C, \qquad x_3 = s_1, \qquad x_4 = s_2.$$

The standard-form minimization problem is

$$\begin{array}{ll} \text{minimize} & c^{\mathsf{T}} x \\ \text{subject to} & Ax = b, \\ & x \geq 0, \end{array}$$

where

$$A = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 2 & 1 & 0 & 1 \end{pmatrix}, \qquad b = \begin{pmatrix} 200 \\ 300 \end{pmatrix}, \qquad c = \begin{pmatrix} -400 \\ -300 \\ 0 \\ 0 \end{pmatrix}.$$

The initial basis and nonbasic tuple are

$$B = (3, 4), \qquad N = (1, 2).$$

Tables and chairs: initial basic feasible solution

For

$$B = (3, 4), \qquad N = (1, 2), \qquad A_B = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

the nonbasic variables are

$$x_1 = x_2 = 0.$$

The basic-variable values satisfy

$$A_B x_B = b. \qquad \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 200 \\ 300 \end{pmatrix} \quad \begin{matrix} x_3 = 200 \\ x_4 = 300 \end{matrix}$$

$A_B^T y = c_B$

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$r_N = c_N - A_N^T y = c_N - 0 = c_N = \begin{pmatrix} -400 \\ -300 \end{pmatrix}$$

Let $k = 1$ be the entering index

Tables and chairs: first simplex iteration

Choose $k = 1$ as the entering index and set

$$x_1 = t.$$

The basic-variable change satisfies

$$A_B \Delta x_B = -a_1.$$

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} (\Delta x_B)_1 \\ (\Delta x_B)_2 \end{pmatrix} = -\begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

Since

$$A_B = I, \qquad a_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix},$$

we obtain

$$\Delta x_B = \begin{pmatrix} -1 \\ -2 \end{pmatrix}.$$

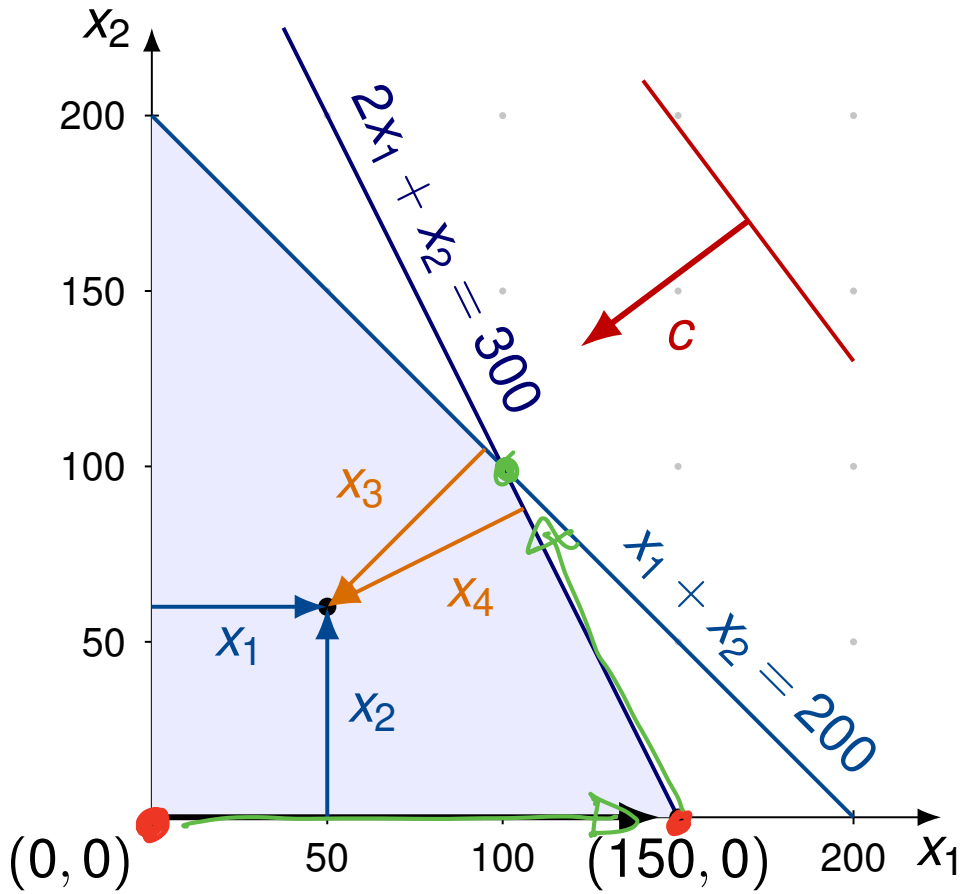
$$\begin{pmatrix} x_3(t) \\ x_4(t) \end{pmatrix} = x_B(t) = \begin{pmatrix} 200 \\ 300 \end{pmatrix} + \begin{pmatrix} -1 \\ -2 \end{pmatrix} t$$

$$t_{max} = \min\left\{ \frac{200}{1}, \frac{300}{2} \right\} = 150$$

Index 4 leaves the basis

Index 1 enters the basis

Tables and chairs: geometric interpretation



The ratio test moves as far as possible in the improving feasible direction.

Tables and chairs: second simplex iteration

At the point

$$x = \begin{pmatrix} 150 \\ 0 \\ 50 \\ 0 \end{pmatrix},$$

use

$$B = (3, 1), \qquad N = (2, 4).$$

The current basic-variable values are

$$x_B = \begin{pmatrix} x_3 \\ x_1 \end{pmatrix} = \begin{pmatrix} 50 \\ 150 \end{pmatrix}.$$

$A_B = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$

$A_B^T y = c_B \quad \begin{pmatrix} 1 & 0 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} 0 \\ -400 \end{pmatrix} \Rightarrow \begin{matrix} y_1 = 0 \\ y_2 = -200 \end{matrix}$

Computer.

Tables and chairs: the optimal solution

At the point

$$x = \begin{pmatrix} 100 \\ 100 \\ 0 \\ 0 \end{pmatrix},$$

the basis and nonbasic tuple are

$$B = (2, 1), \qquad N = (3, 4).$$

$A_B^T y = c_B \quad y \in \mathbb{R}^m$

$r_N = c_N - A_N^T y \quad r_N \geq 0$

Nonnegative reduced costs certify optimality.

When is the problem unbounded?

Suppose that $r_k < 0$ and

$$A_B \Delta x_B = -a_k.$$

If

$$\Delta x_B \geq 0,$$

then increasing $x_k = t$ preserves nonnegativity for every $t \geq 0$.

At the same time,

$$z(t) = c_B^{\mathsf{T}} x_B + r_k t \longrightarrow -\infty.$$

There is no leaving variable and the problem is unbounded.

Finding an initial basic feasible solution

The simplex method requires an initial basic feasible solution.

Assume that $b \geq 0$. Introduce artificial variables $w \geq 0$:

Multiply row by -1 if needed

$$Ax + w = b.$$

$w_i \geq 0$

$e^T w = \sum_{i=1}^{m} w_i = \|w\|_1$

Let e be the vector of ones. The auxiliary linear program

$$\begin{array}{ll} \text{minimize} & e^T w \\ \text{subject to} & Ax + w = b, \\ & x, w \geq 0 \end{array}$$

$\bar{A} = (A \; I) \quad \bar{x} = \begin{pmatrix} x \\ w \end{pmatrix}$

has the initial basic feasible solution $x = 0, w = b$.

- If the optimal value is zero, a feasible solution to the original problem has been found.
- If the optimal value is positive, the original problem is infeasible.

This auxiliary problem is called Phase I of the simplex method.

Main ideas

After this lecture, you should be able to:

- compute reduced costs and use them to test optimality;
- choose an entering variable that improves the objective value;
- determine the feasible change Δx_B;
- determine the maximum feasible step and the leaving variable;
- perform a basis change and continue the simplex method;
- recognize an unbounded linear program.
- Set up a Phase I problem

The simplex method moves along a feasible improving direction from one basic feasible solution to another.

Notation in the compendium

The compendium uses different notation for some quantities in the simplex method:

Compendium	Lectures and formula sheet
β	B
ν	N
$\overline{b}$	x_B
$\overline{a}_k$	$-\Delta x_B$

Indeed, the compendium writes

$$A_\beta \overline{a}_k = a_k, \qquad x_\beta(t) = \overline{b} - \overline{a}_k t,$$

whereas we write

$$A_B \Delta x_B = -a_k, \qquad x_B(t) = x_B + \Delta x_B t.$$

The notation is different, but the simplex iteration is the same.

Question after class

How do we know that the new basis is nonsingular?

We have $A_B \Delta x_B + a_n = 0$. The solution of $(A_B \; a_n)\begin{pmatrix} u \\ v \end{pmatrix} = 0$ is unique up to a scalar, as A_B is nonsingular, i.e. $u = \Delta x_B t, v = t, t \neq 0$

$(\Delta x_B)_p < 0 \Rightarrow u_p \neq 0$. But if the new basis is singular, there must be a solution of $(A_B \; a_n)\begin{pmatrix} \alpha \\ \beta \end{pmatrix} = 0$ with $\alpha_p = 0, \beta = 1$. this contradicts $u_p \neq 0$.